

# A Proof of Bala's Parity Conjecture for the Two-Stack-Sortable Numbers, OEIS A000139

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## Abstract

Let  $a(n) = 2(3n)! / ((2n+1)!(n+1)!)$ , sequence A000139 of the OEIS, which counts two-stack-sortable permutations and rooted non-separable planar maps. In July 2025, P. Bala conjectured that  $a(n)$  is odd precisely when  $n$  is an odd Fibbinary number, that is a term of A022341. We prove it. Legendre's formula turns the parity question into the identity  $s(2n+1) + s(n+1) - s(3n) = 1$  for the binary digit sum  $s$ ; writing  $t$  for the number of trailing 1s of  $n$  and  $c$  for the number of carries in the binary addition  $n + 2n$ , this becomes exactly  $c = t - 1$ . Two short observations finish it:  $c = 0$  if and only if  $n$  has no two adjacent binary 1s, and if  $t \geq 2$  then the trailing block of  $n$  forces  $c \geq t$ , so  $c = t - 1$  is impossible. Hence  $c = t - 1$  holds precisely when  $t = 1$  and  $n$  is Fibbinary, which is precisely membership in A022341.

**2020 MSC.** 11B65, 11A63, 05A10, 11B83.

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## 1 The conjecture

Sequence A000139 of the OEIS [1] is

$$a(n) = \frac{2(3n)!}{(2n+1)!(n+1)!}, \quad n \geq 0,$$

with terms 2, 1, 2, 6, 22, 91, 408, 1938, 9614, .... By a theorem of Zeilberger [2], confirming a conjecture of West,  $a(n)$  counts the two-stack-sortable permutations of length  $n$ ; it also counts rooted non-separable planar maps with  $n+1$  edges, left ternary trees with  $n$  nodes, and fighting fish.

Sequence A022341 is  $4 \cdot \text{A003714}(n) + 1$ , the odd *Fibbinary* numbers, where the Fibbinary numbers A003714 are those integers whose binary expansion contains no two adjacent 1s; A022341 begins 1, 5, 9, 17, 21, 33, 37, 41, 65, .... On 24 July 2025, Peter Bala contributed to A000139:

*Conjecture:  $a(n)$  is odd iff  $n$  is a term of A022341.*

The comment stands unrebutted on the live entry; the page carried the timestamp 22 August 2026.

**Theorem 1.** *For every  $n \geq 0$ , the integer  $a(n) = 2(3n)! / ((2n+1)!(n+1)!)$  is odd if and only if  $n$  is odd and the binary expansion of  $n$  contains no two adjacent 1s; equivalently, if and only if  $n \in A022341$ .*

Throughout,  $s(x)$  is the number of 1s in the binary expansion of  $x$ ,  $v_2$  is the 2-adic valuation,  $t(n)$  is the number of trailing 1s of  $n$  (so  $t(n) = 0$  for even  $n$ ), and  $c(n)$  is the number of carries performed when  $n$  and  $2n$  are added in binary. We call  $n$  *Fibbinary* if it has no two adjacent binary 1s, i.e. if  $n \& 2n = 0$ .

## 2 Reduction to a statement about carries

**Lemma 2.** *For every  $n \geq 0$ ,  $v_2(a(n)) = s(2n+1) + s(n+1) - s(3n) - 1$ .*

*Proof.* Legendre's formula in base 2 reads  $v_2(m!) = m - s(m)$ . Hence

$$\begin{aligned} v_2(a(n)) &= 1 + v_2((3n)!) - v_2((2n+1)!) - v_2((n+1)!) \\ &= 1 + (3n - s(3n)) - (2n+1 - s(2n+1)) - (n+1 - s(n+1)) \\ &= s(2n+1) + s(n+1) - s(3n) - 1. \end{aligned} \quad \square$$

**Lemma 3.** *For every  $n \geq 0$ :*

- (i)  $s(2n+1) = s(n) + 1$ ;
- (ii)  $s(n+1) = s(n) - t(n) + 1$ ;
- (iii)  $s(3n) = 2s(n) - c(n)$ .

*Proof.* (i) Doubling shifts the binary expansion, so  $s(2n) = s(n)$ , and  $2n$  has last digit 0, so adding 1 creates no carry.

(ii) If  $n$  ends in exactly  $t$  ones, adding 1 turns those  $t$  ones into zeros and the next 0 into a 1; the digit sum changes by  $-t + 1$ .

(iii) Kummer's theorem in base 2: the number of carries when adding  $x$  and  $y$  equals  $s(x) + s(y) - s(x+y)$ . With  $x = n$ ,  $y = 2n$  and  $s(2n) = s(n)$  this gives  $c(n) = 2s(n) - s(3n)$ .  $\square$

**Corollary 4.** *For every  $n \geq 0$ ,  $a(n)$  is odd if and only if  $c(n) = t(n) - 1$ .*

*Proof.* By Lemma 2,  $a(n)$  is odd exactly when  $s(2n+1) + s(n+1) - s(3n) = 1$ . Substituting Lemma 3(i)–(iii),

$$(s(n) + 1) + (s(n) - t(n) + 1) - (2s(n) - c(n)) = c(n) - t(n) + 2,$$

so the condition is  $c(n) - t(n) + 2 = 1$ , that is  $c(n) = t(n) - 1$ .  $\square$

### 3 Two facts about the addition $n + 2n$

**Lemma 5.** *For every  $n \geq 0$ ,  $c(n) = 0$  if and only if  $n$  is Fibbinary.*

*Proof.* If  $n$  is Fibbinary then  $n \& 2n = 0$ , so in no binary position do  $n$  and  $2n$  both have a 1; the addition therefore produces no carry at any position and  $c(n) = 0$ . Conversely, if  $n$  is not Fibbinary there is a position  $i$  at which  $n$  and  $2n$  both have a 1 (namely,  $i$  such that  $n$  has 1s in positions  $i$  and  $i - 1$ ), and adding two 1s there produces a carry out of position  $i$  regardless of the carry in; so  $c(n) \geq 1$ .  $\square$

**Lemma 6.** *If  $t(n) \geq 2$  then  $c(n) \geq t(n)$ .*

*Proof.* Write  $t = t(n) \geq 2$ . By definition  $n$  has 1s in positions  $0, 1, \dots, t - 1$  and a 0 in position  $t$ ; consequently  $2n$  has 0 in position 0, 1s in positions  $1, \dots, t$ , and 0 in position  $t + 1$ . Run the schoolbook addition of  $n$  and  $2n$  from the least significant end.

At position 0 the digits are 1 and 0 with no incoming carry: sum 1, no carry out. At position 1 the digits are 1 and 1 (using  $t \geq 2$ ) with incoming carry 0: carry out 1. Inductively, for  $2 \leq i \leq t - 1$  the digits are 1 and 1 with incoming carry 1: sum 1, carry out 1. So positions  $1, \dots, t - 1$  each produce a carry, which is  $t - 1$  carries.

Finally consider position  $t$ . There  $n$  has 0 and  $2n$  has 1, and the incoming carry is 1 (it came out of position  $t - 1$ , which exists because  $t - 1 \geq 1$ ). Thus  $0 + 1 + 1$  produces a carry out of position  $t$  as well. That is one further carry, so  $c(n) \geq (t - 1) + 1 = t$ .  $\square$

### 4 Proof of Theorem 1

*Proof.* By Corollary 4 we must determine for which  $n$  one has  $c(n) = t(n) - 1$ . Since  $c(n) \geq 0$  always, the equation forces  $t(n) \geq 1$ , so  $n$  is odd; and by Lemma 6 it forces  $t(n) \leq 1$ , since  $t(n) \geq 2$  would give  $c(n) \geq t(n) > t(n) - 1$ . Hence  $t(n) = 1$ , and the equation reduces to  $c(n) = 0$ , which by Lemma 5 says that  $n$  is Fibbinary.

Conversely, suppose  $n$  is odd and Fibbinary. Then  $c(n) = 0$  by Lemma 5; and  $n$  has a 1 in position 0, so being Fibbinary it has a 0 in position 1, whence  $t(n) = 1$ . Thus  $c(n) = 0 = t(n) - 1$  and  $a(n)$  is odd.

It remains only to note that the odd Fibbinary numbers are exactly the terms of A022341. Indeed, if  $n$  is odd and Fibbinary then, as just observed, its positions 0 and 1 carry 1 and 0, so  $n = 4m + 1$  with  $m = \lfloor n/4 \rfloor$  Fibbinary; conversely if  $m$  is Fibbinary then  $4m$  has no two adjacent 1s and its two lowest digits are 0, so  $4m + 1$  is Fibbinary and odd. Hence  $\{n : n \text{ odd and Fibbinary}\} = \{4m + 1 : m \in \text{A003714}\} = \text{A022341}$ .  $\square$

### 5 Remarks

*Remark 7* (the boundary case). The conjecture is stated for the entry as indexed, with offset 0 and  $a(0) = 2$ . This matters:  $a(0) = 2$  is even and 0 is *not* a term of A022341, the latter beginning at  $4 \cdot \text{A003714}(1) + 1 = 1$ . The proof covers  $n = 0$  with nothing to spare — there  $t(0) = 0$  and  $c(0) = 0$ , so  $c \neq t - 1$  — and had the sequence been normalised

with  $a(0) = 1$ , as some older presentations of the two-stack-sortable numbers effectively are, the conjecture would have been false at  $n = 0$ .

*Remark 8* (relation to the central binomial case). It is recorded on A003714 that  $\binom{3m}{m}$  is odd if and only if  $m$  is Fibbinary. Since  $a(n) = 2^{\binom{3n}{n}} / ((n+1)(2n+1))$ , Bala's conjecture says that the extra factor  $2 / ((n+1)(2n+1))$  removes exactly the even Fibbinary indices. Our proof does not deduce one statement from the other; it computes  $v_2(a(n))$  directly, and the extra condition “ $n$  odd” emerges as the requirement  $t(n) = 1$  rather than  $t(n) = 0$ .

*Remark 9* (what the criterion is really counting). Corollary 4 is the whole content: the parity of  $a(n)$  is governed by the single integer  $c(n) - t(n)$ , a comparison between the carries in  $n + 2n$  and the trailing block of  $n$ . Lemmas 5 and 6 then say that this quantity equals  $-1$  only in the most degenerate possible situation, namely when there are no carries at all and the trailing block has length one.

## 6 Computational verification

A script (Python 3, standard library) checks the following; all pass.

1. The formula reproduces all 26 terms in the published DATA field of A000139, including  $a(0) = 2$ .
2. A022341, generated as  $4 \cdot \text{A003714}(n) + 1$ , consists of odd Fibbinary numbers, and begins 1, 5, 9, 17, 21, 33, 37, 41, 65, 69, 73, 81.
3. Lemma 2 for  $0 \leq n \leq 400$ .
4. Lemma 3(i) and (ii) for  $n < 20000$ .
5. Corollary 4 for  $0 \leq n \leq 400$ .
6. Lemma 6 for all  $n < 200000$  with  $t(n) \geq 2$ .
7. Lemma 5 for all  $n < 200000$ .
8. The theorem itself for  $0 \leq n \leq 400$ : the indices  $n \leq 200$  with  $a(n)$  odd are exactly 1, 5, 9, 17, 21, 33, 37, 41, 65, 69, 73, 81, 85, 129, 133, 137, 145, 149, 161, 165, 169, which are the terms of A022341 in that range.

Every numeral in this note is an output of that script.

## 7 Concluding remarks

The proof is elementary — Legendre, Kummer, and two paragraphs of carry bookkeeping — and we claim no difficulty for it. It is offered because the conjecture is recent, because the reduction of Corollary 4 seems worth isolating (it converts a parity question about

a product of factorials into a comparison of two binary statistics of  $n$ ), and because the same method applies verbatim to any sequence of the form  $\lambda(\alpha n)!/((\beta n + u)!(\gamma n + v)!)$ .

We have found no earlier proof on the entry or in the literature, but an argument this short could well have been given before, and we would welcome notice of any earlier appearance.

## References

- [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequences A000139, A003714 and A022341; comment of P. Bala, 24 July 2025.
- [2] D. Zeilberger, *A proof of Julian West's conjecture that the number of two-stack-sortable permutations of length  $n$  is  $2(3n)!/((n+1)!(2n+1)!)$* , Discrete Mathematics **102** (1992), 85–93.